

**Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)**  
**Course Name: Applied Mathematics-I Course Code: MA-101L Examination- 2nd periodical**  
**Scheme- CBCS Branch: CSE(AI/ML) Sem: 1<sup>st</sup>**  
**Question Paper for 2nd Periodical Examinations (November, 2023)**

**Time: 2 Hours**

**Max. Marks: 50**

**Note: All questions are compulsory.**

**Course Outcomes:**

*On successful completion of this course the students should be able to:*

CO1. Analyze the convergence/divergence of sequences and series along with error estimates.

CO2. Explain and apply Mean value theorems, Maclaurin's series, Taylor series and their applications.

CO3. Apply integral calculus to evaluate definite and improper integrals, length of arc, surface areas and volume of revolution.

CO4. Identify limits, continuity and partial derivatives of function of several variables.

CO5. Evaluate the maxima and minima by using higher order partial derivatives and Lagrange's method of multipliers.

CO6. Illustrate multiple integrals in different co-ordinates along with their applications in Engineering

**Q.1 (CO4) (a) If  $u = x^2 + y^2$ , then find  $\partial^2 u / \partial x \partial y$ .**

(CO4) (b) State Euler's Theorem for homogenous function of three variables.

(CO4) (c) Evaluate  $\lim$

$(x,y) \rightarrow (0,1) (x+y-1)/(\sqrt{x+y}-1).$

**(CO4) (d) Find the value of  $\int_0^\infty e^{-x} dx$ . (if it converges)**

**(CO6) (e) Find the area of circle by using the double integration?**

**(2×5=10)**

Q.2 (CO3)(a) Evaluate  $\int_{-\infty}^\infty xe^x dx$ . State whether the improper integral converges or diverges. **(5)**

(CO3) (b) Find the value of the improper integral  $\int_{-1}^1 (1/x^2) dx$ . Discuss its convergence. **(5)**

Q.3 (CO4) (a) Evaluate the  $\lim$ , as  $x \rightarrow \infty$ ,  $y \rightarrow 2$  of  $(xy+1)/(x^2+2y^2)$ , if it exists. **(5)**

(CO4) (b) If  $z = \log(u^2 + v)$ ,  $u = e^{x^2+y^2}$ ,  $v = x^2 + y$ , find  $\partial z / \partial x$  and  $\partial z / \partial y$ . **(5)**

Q.4 (CO4) (a) If  $u = x + y + z$ ,  $uv = y + z$ ,  $uvw = z$ , show that  $\partial(x,y,z)/\partial(u,v,w) = u^2v$ . **(5)**

(CO6) (b) Change the order of integration and evaluate the integral  $\int_0^{4\pi} \int (x^2/4a)^{2\sqrt{ax}} dy dx$ . **(5)**

Q.5 (CO5) Find the maximum and minimum value of  $\sin x \sin y \sin(x + y)$ . **(5)**

(CO5) Divide 24 into three parts such that the continued product of the first, square of the second and the cube of the third may be maximum. **(5)**

**(Dr. Kiran Thakur)**